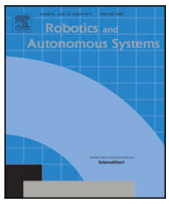
Robotics and Autonomous Systems 185 (2025) 104868



Contents lists available at ScienceDirect

Robotics and Autonomous Systems

journal homepage: www.elsevier.com/locate/robot





Automation of polymer pressing by robotic handling with in-process parameter optimization

Yuki Asano [a,b,*], Kei Okada [c], Shintaro Nakagawa [d], Naoko Yoshie [d], Junichiro Shiomi [a,b,e]

[a] Institute of Engineering Innovation, Graduate School of Engineering, The University of Tokyo, Japan
[b] Department of Mechanical Engineering, Graduate School of Engineering, The University of Tokyo, Japan
[c] Graduate School of Information Science and Technology, The University of Tokyo, Japan
[d] Institute of Industrial Science, The University of Tokyo, Japan
[e] RIKEN Center for Advanced Intelligence Project, Japan

ARTICLE INFO

Dataset link: Robotic polymer pressing (RAS2024) (Original data)

Keywords:
Robotic press process
Lab automation
Autonomous system
Machine learning
Bayesian optimization
Polymer materials informatics

ABSTRACT

In this study, we introduce an autonomous system for polymer pressing that integrates robotic manipulation, specialized equipment, and machine learning optimization. This system aims to significantly reduce lead time and human labor in polymer-materials development. Our approach utilizes an arm-type robot to handle polymer beads and operate a press machine, with process parameters autonomously determined by Bayesian optimization. The keys to this automation are custom-designed press tools that are suitable for robotic handling, such as press plates or fork, a gripper—tool interface with tapered convex and concave parts that enables the handling of multiple tools by a single robot gripper, and an integrated control system that synchronizes the robot with the press machine. Additionally, we implement a closed-loop process that incorporates image processing for pressed-polymer recognition and Bayesian optimization for continuous parameter refinement, with an evaluation function that considers polymer-film thickness and press times. Verification experiments demonstrate the capability of the system to autonomously execute pressing operations and effectively propose optimized press parameters.

1. Introduction

In the field of materials science and development, the research and use of data-driven approaches that utilize materials informatics (MI), combining data generation through simulations and experiments and black-box learning with machine learning (ML), for the development of new materials with innovative functions have been accelerated in both academia and industry [1]. In the analysis of big data generated in multidimensional spaces, ML is necessary to arrive at solutions beyond human thinking capabilities. This has the potential to change the traditional approach to materials research, which is based on academic knowledge and human experience.

To implement MI based on experimental data, a large amount of data must be generated during actual experiments. This requires the automation of laboratory work (lab automation), for which automated robotic systems [2,3] are an important technology. Robotic automation in scientific experiments enables the generation of larger amounts of and more reproducible data through high-throughput experiments than humans are capable of producing. Although many research studies on experiment automation [4–8] have been conducted, rapid developments have recently been made combining experiment automation and ML, particularly in the fields of biology [9–12], chemistry [13,14], and materials science [15–17].

The challenge in materials lab automation is to produce large, well-defined, and reproducible experimental data, including input process parameters and output material properties. Therefore, automation of the production and characterization processes is important. However, materials development often involves many different types of equipment and instruments for each process, such as synthesis, molding, and characterization. To address this issue, it is necessary to develop a system that integrates robots with experimental equipment. System integration enables the cooperation of multiple control programs that configure the system to work together to ensure that the appropriate function is executed at the appropriate time. Of particular interest in helping to accomplish this purpose is the use of a multi-degree-of-freedom (DOF) robot arm with a DOF equivalent to that of a human arm, which is effective for automating devices that are usually operated by humans and require operation input by hand from multiple directions.

* Corresponding author.
E-mail addresses: yasano@g.ecc.u-tokyo.ac.jp (Y. Asano), k-okada@jsk.t.u-tokyo.ac.jp (K. Okada), snaka@iis.u-tokyo.ac.jp (S. Nakagawa), yoshie@iis.u-tokyo.ac.jp (N. Yoshie), shiomi@photon.t.u-tokyo.ac.jp (J. Shiomi).

https://doi.org/10.1016/j.robot.2024.104868
Received 18 May 2024; Received in revised form 29 October 2024; Accepted 17 November 2024
Available online 21 November 2024
0921-8890/© 2024 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

Autonomous materials development is enabled by automating the processes of synthesis, molding, and property evaluation, and then by building a closed loop for the development, where the input for the next cycle are the parameters that maximize the performance as predicted through ML, by which the correlation between the process parameters and properties is learned [18,19].

Here, we focus on lab automation for the development of polymer materials [20,21]. Polymers are used in a wide range of products, from daily necessities to industrial goods, and are indispensable in our daily lives. They are generally lighter than metals, flexible and moldable, and have the versatility to change their properties depending on the processing. Polymers can be reinforced by compounding with fibers or fillers to enhance a variety of properties, including mechanical [22], electrical [23], and microwave absorption [24] properties. Regarding MI for polymers, the current state of the art is mostly based on simulations. For these, open-source databases outlining different properties calculated using molecular dynamics methods have been constructed [25]. The development of efficient computational methods by applying data extensions commonly used in computer vision and natural-language processing to polymer MI and the improvement of the accuracy of properties prediction, particularly for polyetherimide, have been reported [26]. Conversely, with regard to MI by experimentation that includes the process parameters, an ML model for the reaction process of a binary copolymer by flow reactor has been constructed [27], and a high-throughput experimental system for generating copolymers from comonomers has been reported [28]. For synthesis, which is the initial step of materials development, the design and creation of automated synthesis platforms [29] and continuous flow systems [30] have been proposed. However, to the best of our knowledge, for the processes after synthesis, such as film molding and property measurement, reliable and effective solutions have yet to be established because of the complexity of these processes and the difficulty of handling polymer materials, which changes these forms to solid granules or deformed film in the processes.

In this study, we report an automated system for the press process used for molding products. For this complex development, we constructed an automation system based on robotic handling and an integrated system between the robot control and press process equipment. This automated press-equipment operation requires operational inputs from multiple directions and robot/equipment control coordination. To achieve an autonomous experimental loop, the robot press system was integrated with an ML-based parameter search system that uses Bayesian optimization (BO), by which the press process parameters (input) and polymer material properties (output) are optimized.

2. Press process in polymer-material development

An overview of polymer-material development and the press process is illustrated in Fig. 1. Polymer-material development can be roughly divided into three processes: synthesis, molding, and property characterization. The relationship between the input process parameters and output material properties is analyzed to determine the parameters for the next trial, and by iterating these processes, the material properties are optimized.

Among the aforementioned processes in the experimental loop, this study focuses on the molding process, particularly the press molding process. Press molding is commonly used to fabricate films through the heat pressing and cold pressing of polymer granules, following requirements in terms of property measurements or applications. Presently, the press process is performed by humans, and there have only been a few specialized equipment and studies to automate it. Press processes are time-consuming because they involve the melting and spreading of the polymer inside the press plate and the cooling of the polymer to a fixed shape, both of which are slow. Therefore, a significant amount of time is usually required for the entire polymer development. In addition, the process requires the delicate handling of granules and deformed film, making it challenging to reproduce. In further developments using MI, a large number and different types of polymers will have to be pressed with appropriate press parameters for each polymer, where the human factor can become a crucial limitation. Therefore, the automation of the press process, which will enable high-throughput and reproducible data generation, is highly important for accelerated materials development.

3. Methods

3.1. Overview

The overall robotic system for the automated polymer press process is shown in Fig. 2. Press machines and several tools on the workstand used to perform the press process were positioned around the robot arm. In addition to the automation of the press operation, the integration of automatic parameter exploration using the evaluated data and ML for the next trial is essential to form an autonomous loop of the process. In summary, for the automation of the polymer press process, we propose an autonomous system for press operation that uses a robot arm and parameter exploration using BO.

3.2. Robot arm

For this study, to automate the press molding process, a robotic system that uses robot arms was implemented. Robot arms are suitable for automating complicated tasks performed by humans [2,3], including laboratory work, which is the focus of this study, and fit into the environment of humans. Thus, the advantage of their use is that commercially available laboratory equipment such as press machines can be automated without special modifications. In addition, this approach of adoptable automation makes the system more flexible than embedding automated parts into instruments for a specific process. This generally makes it easy to connect the system to another process and rearrange the layout of the system via expansion and future changes in the development environment.

Panda (Franka Emika) [31], which is a collaborative 7DOF robot arm, was adopted for the proposed system (bottom left of Fig. 2). Its payload is 3 [kg], and its maximum reach is 855 [mm]. A camera (Intel RealSense D435i) was attached to the robot hand to obtain images of the environment.

3.3. Press machine

The press machine (Imoto Machinery Co., Ltd) is a combined equipment composed of a heat press (HP) machine and a cold press (CP) machine (top center of Fig. 2). The machine is controlled by a programmable logic controller (PLC). Control commands for the start/stop of press operations and press settings can be sent to the machine. In addition to the usual operation of the machine via its touch panel, control commands from a computer can be used.

The parameters of the press machine that can be controlled are as follows: four parameters for the HP including the press force F^{HP}[kN], press temperature of heating plates C^{HP}[°C] (for each upper and lower heater), and press time T^{HP}[s]; and two parameters for the CP including the press force F^{CP}[kN] and press time T^{CP}[s].

3.4. Customized press tools for the robotic system

In general, it is difficult for a robot arm to handle human tools in the same manner as humans. Therefore, it is necessary to customize and design tools that can be handled easily by a robot arm. Regarding to the press operation, the press plate is an important tool for robotic automation. A conventional press plate, which consists of a base plate for placing polymers, an intermediate plate for creating a molding gap, and an upper plate for pressing polymers, is not easy for robots to handle. Therefore, we created a fork-shaped tool for lifting and

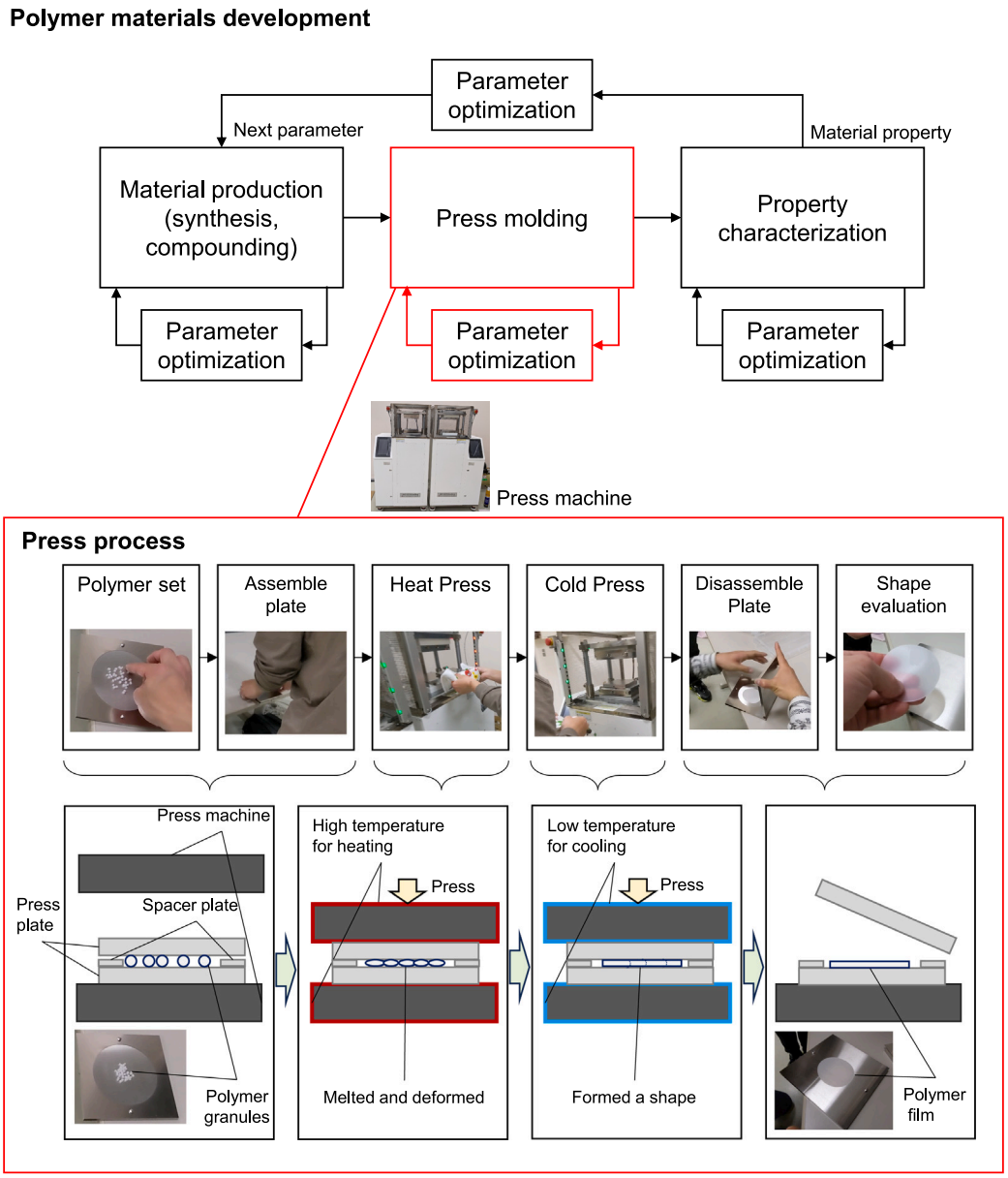


Fig. 1. Overview of polymer-materials development, including press process. Red parts indicate focus of this study. Top: Processes under polymer-materials development in closed loops of trials. The parameters for each process are optimized to obtain better results through iterative trial and error. Bottom: Process flow of press process. Polymer granules are pressed to fabricate films using a press machine composed of a heat press and cold press. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

transporting the press plates (top right of Fig. 2). Each plate has grooves on its underside to secure it onto the fork as it is transported.

In addition to the press plates, several other useful tools were developed (bottom right of Fig. 2), including a feeder cup for manipulating polymer granules onto the press plates, and a feeder support plate for preventing polymer granules from scattering when the granules are placed onto the press plates. These tools are positioned on the designated tool holders on the workstand.

3.5. Gripper–tool interface

A gripper–tool interface, which is a mechanical interface on the hand and tool sides (bottom center of Fig. 2), was developed to enable the use of multiple tools with only one robot arm. This interface is composed of tapered mating convex and concave parts that fit each other, enabling the gripper to grasp a tool. Two tapered convex parts are on the robot-hand side, whereas two tapered concave parts are on the tool side. This interface allows positional errors between the hand and tools to some extent. It is also effective at restricting the rotation of heavy objects such as press plates during manipulation.

3.6. Integrated system of robot-equipment-ML

To autonomize the press process, several software components must be integrated into a single system (Fig. 3). The control system for the robot arm was implemented using Robot Operating System (ROS) framework [32], which is an open-source robot middleware. In contrast, the press machine equipment was controlled by the PLC framework. A bridge software between ROS and PLC was implemented for their communication. Using this bridge, data and control commands were exchanged via ROS message format, allowing the user to control the press machine in the same manner as the robot.

In addition, we implemented another bridge using ROS framework to enable communication between the robot system and ML library

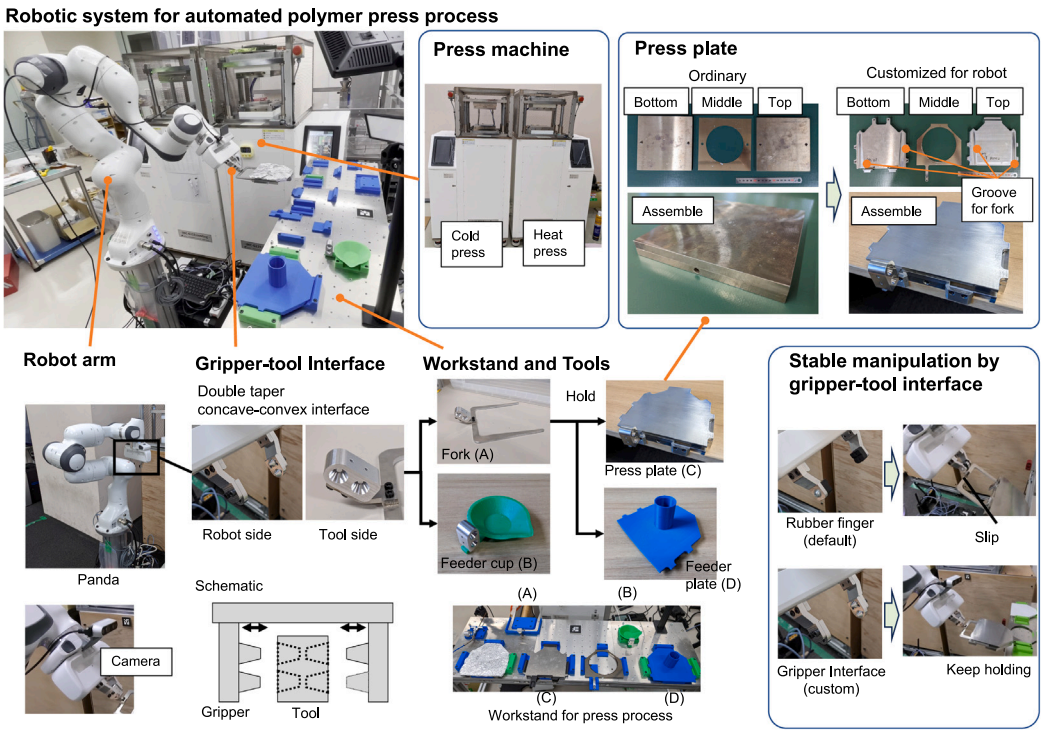


Fig. 2. Automation environment of the robotic system for polymer press process. The robot arm (3.2), press machine (3.3), press plates and tools (3.4), and gripper–tool interface (3.5) are shown. The robot arm is positioned at center of environment. The press machine and workstand are positioned to surround the robot. Gripper–tool interface enables stable manipulation of press plate by preventing plate slippage due to its weight.

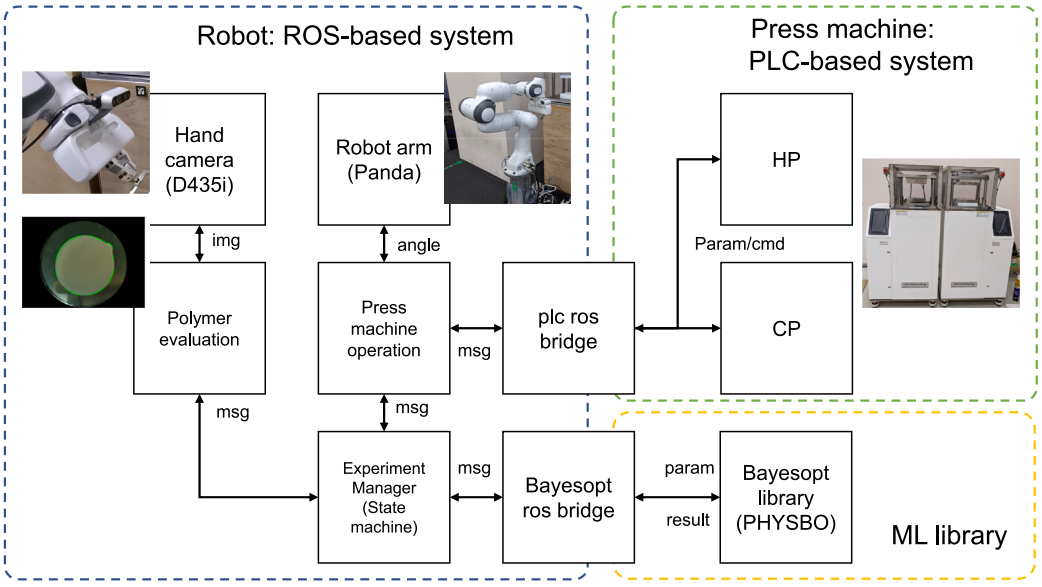


Fig. 3. Integrated system between ROS, PLC, and ML library. Integration of these components into a single system is necessary to achieve seamless process automation, including in-process optimization of parameters, without manual human operation. State machine manages the experiment procedures and sends commands to each software node to start its function at the appropriate timing. Data, parameters, and commands are communicated among components.

PHYSBO. By enabling the operations of PHYSBO through the ROS message, we can add the resulting data to the library and obtain the selected parameters from the library during the robotic process without human intervention.

3.7. Polymer material

The polymer material used in this study was high-density polyethylene (HDPE). The reference value of its density is 0.956 [g/cm^3]. Polyethylene is a common polymer material that is easy to mold and can be commercially obtained. Its melting temperature is approximately 130 [°C], and its molding temperature is approximately 180–270 [°C].

3.8. Polymer thickness estimation through polymer film recognition

Numerical evaluation is necessary for an autonomous system to determine whether the result is better or worse than a certain standard for

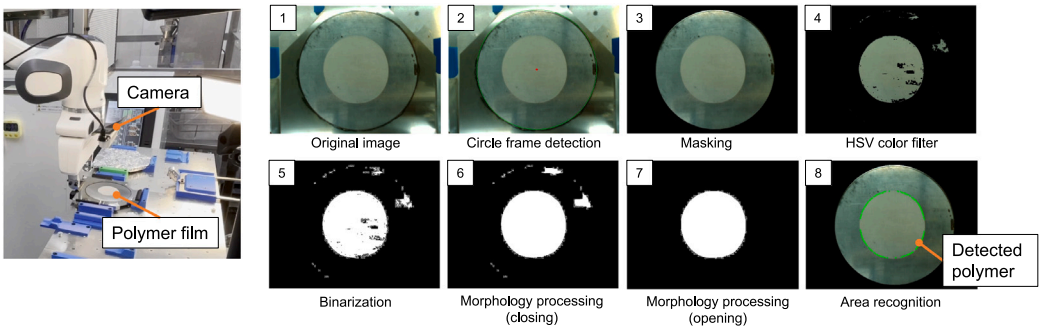


Fig. 4. Thickness estimation procedure using image recognition for the pressed film polymer. 1. Raw image of the molded polymer. 2. Circular inner frame of the middle plate was detected by the Hough circle transform. 3. Outer side of the circular frame was masked to focus on the region of the polymer film. 4. HSV color filter was applied to extract the polymer film. 5. Binarization was applied to simplify the image for additional image processing. 6. Hole (black part) was filled by the closing processing of the morphological operation. 7. Nonpolymer region (outside white part) was removed by the opening processing of the morphological operation. 8. Remaining area was recognized as the polymer film and used for thickness calculation.

comparison. In the press process for polymers, the thickness, circularity, color, and uniformity are important factors because they affect the downstream property characterization (Fig. 1) or application. In this study, we developed a method for estimating the film thickness using image processing. Through this method, the average film thickness can be estimated from an overall view of the entire film. The film thickness can influence the property calculation and must be tuned to fit the requirements of the instruments used for property characterization. The thickness is affected by the conditions for sufficient melting and pressing.

The thickness of polymer film is calculated as follows:

$$St = V \tag{1}$$

$$t = \frac{V}{S} = \frac{1}{S}\frac{m}{d} \tag{2}$$

where t [mm] is the thickness, S [mm^2] is the surface area, V [mm^3] is the volume, d [g/cm^3] is the density, and m [g] is the mass. Based on these relationships, when the mass m and density d of a polymer are assumed to be known, the thickness t can be obtained using the area S. Therefore, we implemented image processing to determine the surface area of the polymer film to estimate its thickness. With the camera attached to the robot, the top-view shape of the molded polymer film can be obtained after the press process.

To calculate the polymer area S, the following equation was used considering the relationship between the number of pixels in the image and the actual radius of the circular frame.

$$S = n_{poly} S_{pixel} = n_{poly}\left(\frac{r_{fr}}{n_{fr}}\right)^2 \tag{3}$$

Here, n_{poly} is the number of pixels for the recognized polymer film, S_{pixel} is the actual area corresponding to 1 pixel, r_{fr} is the actual radius of the frame (50 mm), and n_{fr} is the number of pixels for the radius of the frame. Each number of pixels can be obtained using functions in OpenCV library through the image processing method described below.

Fig. 4 shows the procedure of the image processing and recognition result for a sample polymer. The estimated thickness of the sample was 0.454 [mm]. The actual thickness, which is the average of ten points measured using a thickness gauge, was 0.439 [mm] (corresponding to an error of 3.4 %).

3.9. Parameter exploration using ML

To understand the relationship between the input material parameters and output evaluation values (physical properties), it is necessary to obtain input/output datasets from real trials. This relationship often has an underlying black-box function. In addition, actual trials are time-consuming. Thus, it is desirable to find the appropriate parameters with as few trials as possible using an ML method.

BO is a hyperparameter optimization method using the Gaussian process and stochastic prediction for the exploration of better parameters. Applications of BO are often observed in the ML field to optimize systems defined with black-box functions [33]. BO is particularly suitable when the cost of data acquisition is high, such as in real-world trials that require several hours for a trial, experiments that require expensive specimen such that only a limited number of specimen can be prepared, or complex simulations that require several weeks for a simulation. As a comparison with other optimizations such as metaheuristic, BO is a candidate strategy to obtain better results from smaller number of evaluations. Another advantage of BO is that it is effective for the objective function, which has multiple local maximum owing to its working principle of adjustment of exploration and exploitation in parameter selection. A brief formulation of BO is provided in Appendix A. In this study, a parameter exploration system that employs BO was implemented using the open-source BO library PHYSBO [34].

3.10. Evaluation function for press process

To optimize the parameters of the press process, devising an evaluation function is effective. If an evaluation function is used as the objective function of BO, optimized parameters can be obtained to achieve better results. The following evaluation function, which evaluates a pressed polymer, was devised for the press process:

$$y = \begin{cases} 0 & (t > t_{thre}) \\ 1 - w_T \frac{T_{pre}+T_{press}}{T_{max}} - w_t \frac{t}{t_{thre}} & (t \le t_{thre}) \\ (w_t + w_T = 1) & \end{cases} \tag{4}$$

where T_{pre} is the press time for the preheating process, T_{press} is the press time for the main press process, T_{max} is the maximum heating time (upper limit in the parameter space), t is the polymer thickness, t_{thre} is the thickness threshold, w_t is the weight of the thickness parameter, and w_T is the weight of the time parameter. In this study, we used $w_t = w_T = 0.5$, $t_{thre} = 0.3$ [mm], $T_{max} = 800$ [s].

Eq. (4) yields high values when the press time is shorter and the thickness condition is satisfied. This function includes terms for the press time and thickness, in addition to the thickness threshold, and yields a low value if the polymer film is thick. We devised this function by assuming situations that asymmetrically penalize the degree of deviation from the expected thickness (e.g., some property measurement instruments require film thickness to be less than a certain value). This function enables the user to incorporate various demands and purposes for the press process. In this case, these conditions are considered effective because an efficient press process can be achieved by reducing the press time while simultaneously satisfying the condition for the necessary thickness.

3.11. Parameter definition

In this study, 10 parameters of the press process were considered by dividing the entire press process into three steps: the preheat process for melting (HP), the main heat-press process (HP), and the cold-press process (CP). The press times for the two presses using the HP were set as the variable parameters $x_i = (T^{HP}_{pre}, T^{HP}_{press})_i$, whereas the others were set as fixed parameters $F^{HP}_{pre} = 0.05$ [kN], $F^{HP}_{press} = F^{CP} = 0.3$ [kN], $C^{HP,u}_{pre} = C^{HP,l}_{pre} = C^{HP,u}_{press} = C^{HP,l}_{press} = 180$ [°C], $T^{CP} = 200$ [s].

To allow for the stable molding of the polyethylene material, the press force and temperature parameters were fixed. Excessive press force and temperature make the plate disassembly difficult because the melted polymer sticks to the upper and lower press plates. Therefore, the focus of this study is to confirm the effects of changes in press time on the molded polymer and to verify the system based on this definition.

4. Results

4.1. Configuration of autonomous robotic press process

With the parameters defined earlier, we executed the autonomous robotic press process, in which the automated press process, evaluation, and parameter exploration via BO were repeatedly run by the robotic system. Fig. 5 illustrates the loop of this process.

The parameter dimension, including the preheating press time T^{HP}_{pre} and the main heat-press time T^{HP}_{press}, is 2, and thus, $x_i = (T^{HP}_{pre}, T^{HP}_{press})_i$. Twenty parameter candidates were produced in total. At the beginning of each process, a cup of polyethylene (1.0 g) was weighed and placed on the workstand by an operator. The robot arm proceeded with the press process to produce a molded polymer, which was captured by the camera. The recognized polymers were evaluated using the evaluation function, Eq. (4). These processes were repeated ten times to create a dataset, which includes the press process parameters and evaluation results. The parameters were selected by random search (pseudo-random) for repetitions 1–3 and by BO for repetitions 4–10. This was terminated after 10 repetitions of the processes, and the results were evaluated.

4.2. Result of autonomous robotic press process

A mapping of the evaluation results and process parameters is shown in Fig. 6. The maximum value obtained in the evaluation was 0.388 for process 3 with the parameter $(T^{HP}_{pre}, T^{HP}_{press}) = (0, 200)$, where the thickness was 0.292 [mm]. As a result, we could confirm that the parameter exploration worked well because the parameters with higher evaluation values were selected within the initial 10 times for all 20 processes.

Lower evaluation results at process indices 4 and 7 correspond to the press parameters and results that were over the threshold defined by Eq. (4) due to shorter press times. After these process indies, BO searched other regions to find better evaluation results.

5. Discussion

5.1. Verification of parameter exploration

To verify the parameter exploration system, a verification experiment on molded polymers was conducted by a human experimenter. The experimenter molded 20 polymer samples using different process parameters and evaluated their thicknesses using the proposed image-recognition method. The left side of Fig. 7 shows a mapping of the original estimated thickness values.

The other evaluation function, Eq. (5), which provides a higher evaluation value the thinner the fabricated polymer, is defined for comparison:

$$y = 1 - t \tag{5}$$

The values obtained using the evaluation functions are shown at the center and on the right side of Fig. 7. For Eq. (5), high evaluation values are observed in the region with longer press times. This result is reasonable because the polymer spreads thinly with a long press time; however, it is not time-efficient. In contrast, for Eq. (4), the region that clears the threshold and indicates shorter press times corresponds to high evaluation values. Low evaluation values are observed in the regions corresponding to long press times and thicknesses greater than the threshold due to shorter press times. These results can be considered reference data for the autonomous robotic press process described in Section 4 because the same parameters and functions were used.

Thus, Eq. (4) is preferable because it enables a time-efficient press process. In addition, the distribution of the evaluation results resembles that of the results of the autonomous robotic press process. This suggests that the autonomous robotic press process was successfully completed, with no notable differences from the process conducted by a human experimenter.

5.2. Order of optimal parameter

The order in which the optimal values were obtained was evaluated to further verify the parameter exploration system. To conduct this verification without producing film samples again, we used the known input/output data set (Fig. 7). The procedure was as follows: input parameters were selected in the same manner as in Section 4, after which the corresponding outputs were obtained from the data set, and finally, the order of selection of the optimal value was obtained. In this manner, one trial included searching for all 20 parameters and obtaining the corresponding evaluation values. We conducted ten trials and calculated the average order for the selection of the optimal value. The results are presented in Table 1. The average orders at which the optimal values were obtained were 8.6 times for Eq. (5) and 6.9 times for Eq. (4), which were faster than half of the entire process. Therefore, using BO for parameter searching allows for a more efficient search than searching randomly from the parameter space.

5.3. Limitations and future works

Future work will include improving the robustness of the system to enable the creation of a large number of samples by continuously operating the system for a long time. At present, when the robotic process fails because of errors, a human operator should recover and restart the system. Thus, robust robot manipulation, failure detection, and recovery by the system itself are proposed features for future versions of the system. In addition, the system must be expanded to cover not only the press process, but also the entire process of polymer development. Presently, polymer-development steps before and after the press process, such as weighing polymer granules before pressing and retrieving the polymer film after pressing, are conducted by a human operator. The robotic automation of measurement processes to characterize the fabricated polymers is also important for making the system more practical for real material development. Parameter optimization between the input material parameters and output physical properties is expected to be pursued based on the needs of researchers in materials development.

5.4. Contribution of this system

To the best of our knowledge, this system is an early attempt to automate the development of polymers, especially the press process. We propose a system integrating a robot arm, equipment, and ML, autonomized through the implementation and integration of software components, for the robotic press process as a concrete example. Demonstration of the robotic press process verifies the autonomous framework proposed in this study.

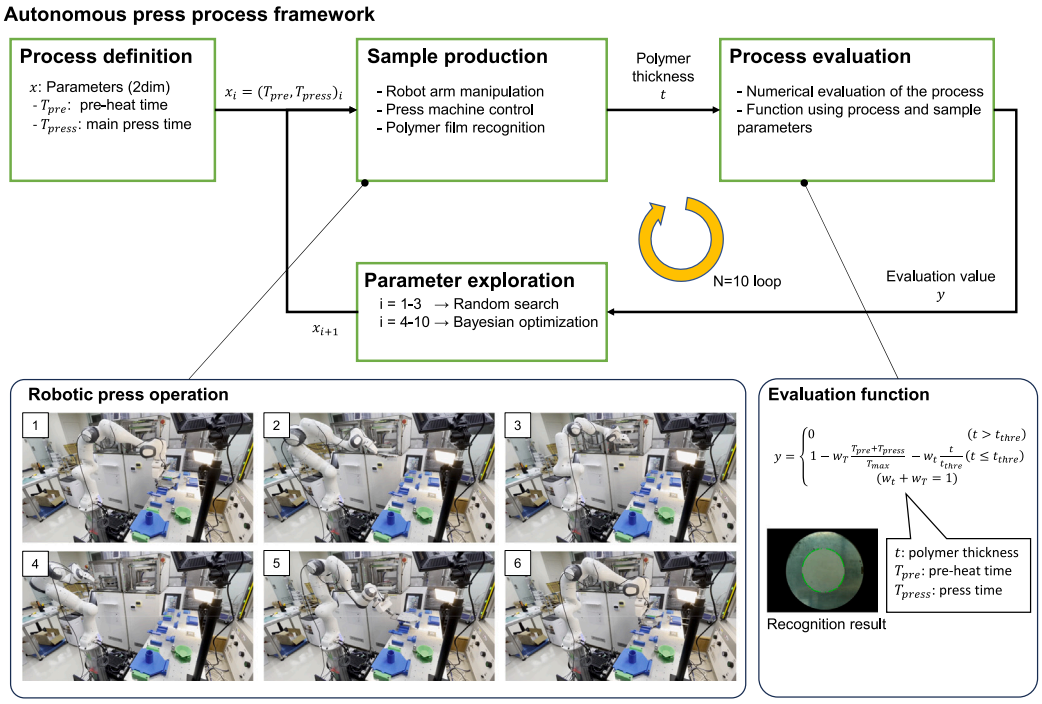


Fig. 5. Framework of the autonomous robotic press process proposed in this study. This is a practical example of a closed-loop of the press molding process illustrated in Fig. 1. As the sample production component, the robot arm executes the press process to produce a polymer film. As the process evaluation component, the molded film and the press process are evaluated using Eq. (4). As the parameter exploration component, press parameters are proposed for the next trial.

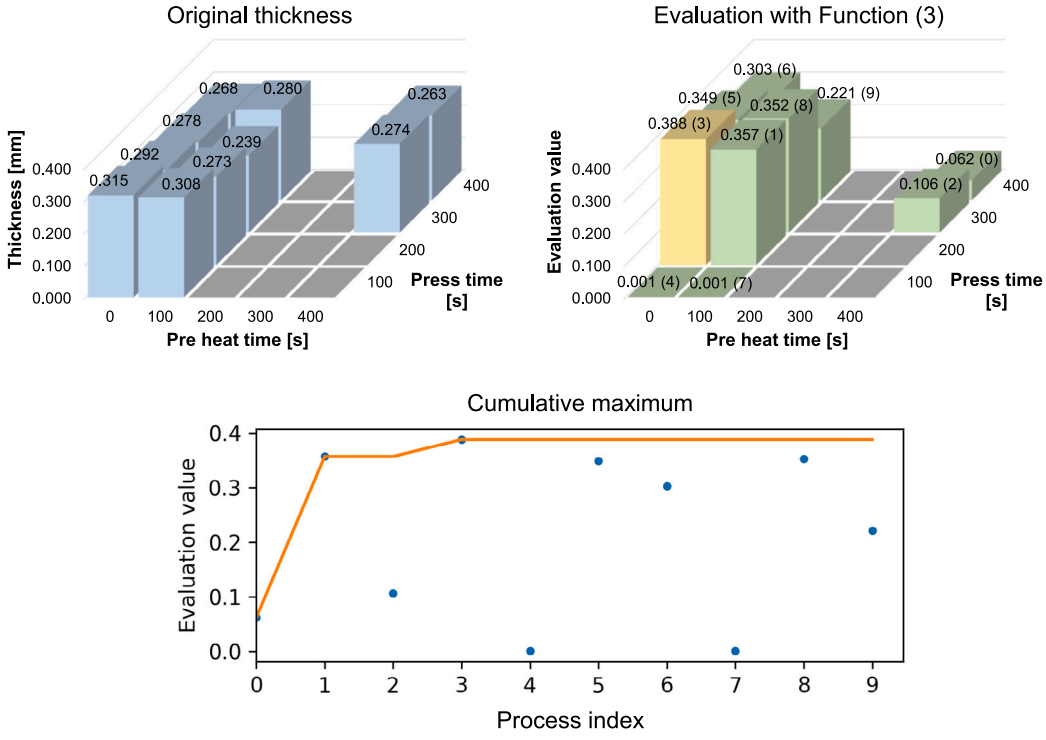


Fig. 6. Results of the autonomous robotic press process. Left: Polymer thicknesses [mm] (estimated by the image processing (3.8)). Right: Mapping of evaluation results and process parameters of robotic press process. Numbers inside brackets are the process indices. Bottom: Cumulative maximum of the evaluation values.

Table 1
Order of obtaining optimal values.

	Trial 1	2	3	4	5	6	7	8	9	10	Average
Eq. (5)	11	14	8	8	8	4	3	11	10	9	8.6
Eq. (4)	6	5	1	15	10	3	5	9	10	5	6.9

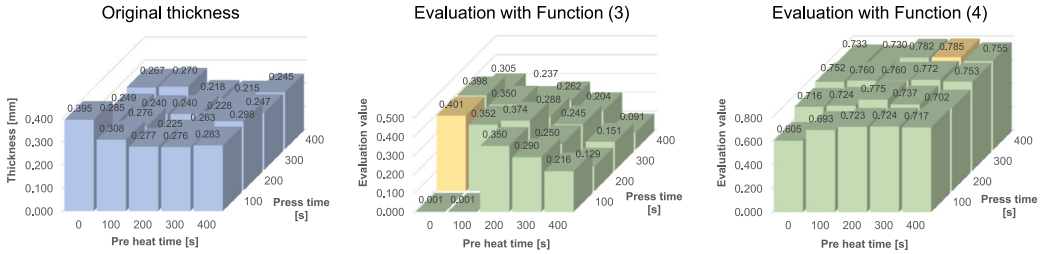


Fig. 7. Results of verification experiment by a human experimenter. Left: Polymer thicknesses [mm] (estimated by the image processing (3.8)). Center: Evaluation results based on Eq. (4). Yellow bar represents the highest evaluation value. Right: Evaluation results based on Eq. (5). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

A practical contribution of this system is that with this approach, time-efficient parameters for the press process that satisfy the thickness restriction for the post-process measurement can be found more easily. In addition, this approach helps establish numerical criteria for the press process that, thus far, have been dependent on human experience. Even in situations in which unfamiliar polymers are pressed, the system helps find effective press parameters depending on their characteristics. This leads to a reduction in the waiting time of the press and enables the production of more polymer products within the same period.

6. Conclusion

In this study, we describe an autonomous system for the polymer press process that integrates a robot arm, press machine, and ML. The proposed system, which was implemented based on the framework of autonomous materials development, consists of an automated robotic process and parameter proposal using ML. Customized tools for the robotic press process were developed. To integrate the different software components, bridges between the robot, press machine, and ML library were implemented. Based on a demonstration of the system in the pressing of polyethylene, the system can successfully and autonomously proceed with and cycle the process. This system is an early trial to automate the press process in polymer development, which is considered difficult to automate because of its complex processes and difficulty in material handling. These results are expected to contribute toward the acceleration of materials science and development. We will continue to develop robotic systems for the discovery of novel polymer materials and to explore the fusion of robotics and materials development.

CRediT authorship contribution statement

Yuki Asano: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Kei Okada:** Supervision, Software, Project administration, Methodology, Funding acquisition, Conceptualization. **Shintaro Nakagawa:** Resources. **Naoko Yoshie:** Project administration, Funding acquisition. **Junichiro Shiomi:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Yuki Asano, Kei Okada, Junichiro Shiomi has patent #P2024-35708A pending to JPO. Yuki Asano, Kei Okada, Junichiro Shiomi has patent #P2024-35709A pending to JPO. Yuki Asano, Kei Okada, Junichiro Shiomi has patent #2022P01422 pending to JPO. Yuki Asano, Kei Okada, Junichiro Shiomi has patent #PCT/JP2023/031471 pending to WIPO. Yuki Asano, Kei Okada, Junichiro Shiomi has patent #PCT/JP2023/031472 pending to WIPO. Yuki Asano, Kei Okada, Junichiro Shiomi has patent #PCT/JP2024/014791 pending to WIPO. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We thank Dr. Koki Kitai for practical advice and discussions related to machine learning techniques. This research was supported by DAIKIN Industries, Ltd., Japan, and JSPS KAKENHI, Japan Grant Number 23K03771.

Appendix A. Bayesian optimization

A.1. Principle

Bayesian optimization is a strong application of the Gaussian process to determine the maximum value of objective functions that are expensive to evaluate (Fig. A.8). To achieve sample-efficient optimization, BO uses an acquisition function based on the Gaussian process as a surrogate model of the objective function. The next evaluation is proposed at the parameter that maximizes the acquisition function and then the acquisition function is updated through the Gaussian process for the next trial. The decision is incorporating an automatic trade-off between exploration (searching from areas of high uncertainty) and exploitation (searching from areas likely to obtain improvement based on the current observation).

A.2. Formulation

The formulation of Bayesian optimization is as follows. If the input parameter space is denoted as $\chi = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_n\}$ (explanatory variables), the output is denoted as $\boldsymbol{y} = \{y_1, y_2, \ldots, y_n\}$ (objective function), and the experimental error is denoted as ϵ. The black-box function f is then defined as follows:

$$y_i = f(\boldsymbol{x}_i) + \varepsilon_i \tag{A.1}$$

We consider the input $\boldsymbol{x}^*$ to maximize the function:

$$\boldsymbol{x}^* = \arg\max_{x \in \chi} f(\boldsymbol{x}) \tag{A.2}$$

Although the function f should be maximized directly, it is assumed to be difficult to obtain a sufficient dataset owing to high experimental costs. Here, through the introduction of a Gaussian process as a statistical model of f, the model can be trained using a relatively small number of data points.

A.3. Acquisition function

For the Gaussian process, an acquisition function α can be defined such that it is calculated from the predicted value and the variance.

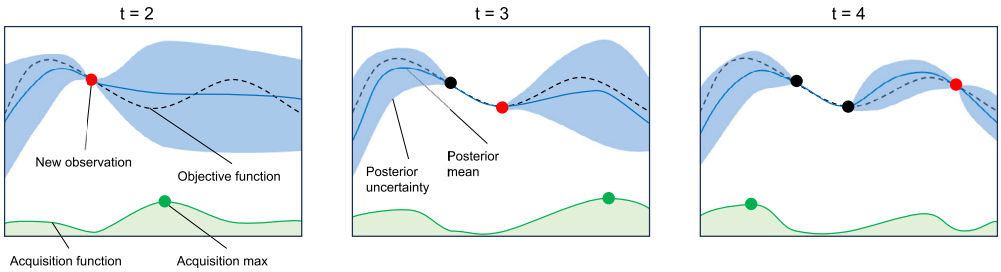


Fig. A.8. Conceptual illustration of BO. Unknown objective function is predicted through Gaussian process regression. The next observation is determined at the point where the highest value of acquisition function. In each optimization process, posterior mean and uncertainty are updated after the observation (evaluation of the trial), and then, the acquisition function is calculated to determine parameters for the next trial. After several iterations of these optimization steps, the maximum value is obtained sample-efficiently.

Acquisition functions are defined to have high values corresponds to potentially high values of the objective function. The maximum value of f is estimated through the maximization of the acquisition function:

$$\boldsymbol{x}^* = \arg\max_{x \in \chi} \alpha(\boldsymbol{x}) \tag{A.3}$$

In this study, we used probability of improvement (PI), which is a search policy that evaluates points with a high probability of improving the objective function value, as the acquisition function:

$$\alpha_{PI}(\boldsymbol{x}) = \mathbb{P}(f(\boldsymbol{x}) \geq y_{max}) \tag{A.4}$$

$$= \Phi(Z), \quad Z = \frac{\mu_c(\boldsymbol{x}) - y_{max}}{\sigma_c(\boldsymbol{x})} \tag{A.5}$$

where Φ is a cumulative distribution function, μ_c is the mean of predicted values, σ_c is the variance, and y_{max} is the maximum evaluation value for each $\boldsymbol{x}$ used. Other acquisition functions include expected improvement (EI), upper confidence bound (UCB), and Thompson sampling (TS).

Appendix B. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.robot.2024.104868.

Data availability

Data will be made available on request.

Robotic polymer pressing (RAS2024) (Original data) (Mendeley Data)

References

[1] S. Ju, J. Shiomi, Materials informatics for heat transfer: Recent progresses and perspectives, Nanoscale Microscale Thermophys. Eng. 23 (2) (2019) 157–172, http://dx.doi.org/10.1080/15567265.2019.1576816.

[2] N. Benhabib, V. Padois, D. Daney, Securing industrial operators with collaborative robots: Simulation and experimental validation for a carpentry task, in: 2020 IEEE International Conference on Robotics and Automation, ICRA, 2020, pp. 7128–7134.

[3] M. Schwarz, A. Milan, C. Lenz, A. Munoz, A.S. Periyasamy, M. Schreiber, S. Schuller, S. Behnke, Nimbro picking: Versatile part handling for warehouse automation, in: 2017 IEEE International Conference on Robotics and Automation, ICRA, 2017, pp. 3032–3039.

[4] F. Cardoso, A distributed system for laboratory process automation, IEEE Trans. Nucl. Sci. 36 (5) (1989) 1624–1627, http://dx.doi.org/10.1109/23.41105.

[5] M. Kaluz, J. Garcia-Zubia, M. Fikar, L. Cirka, A flexible and configurable architecture for automatic control remote laboratories, IEEE Trans. Learn. Technol. 8 (3) (2015) 299–310, http://dx.doi.org/10.1109/TLT.2015.2389251.

[6] P. Najmabadi, A. Goldenberg, A. Emili, A scalable robotic-based laboratory automation system for medium-sized biotechnology laboratories, in: IEEE International Conference on Automation Science and Engineering, 2005., 2005, pp. 166–171, http://dx.doi.org/10.1109/COASE.2005.1506763.

[7] B. Gode, S. Holzmuller-Laue, K. Rimane, M.-Y. Chow, N. Stoll, Laboratory information management systems - an approach as an integration platform within flexible laboratory automation for application in life sciences, in: 2007 IEEE International Conference on Automation Science and Engineering, 2007, pp. 841–845, http://dx.doi.org/10.1109/COASE.2007.4341780.

[8] B.P. MacLeod, F.G.L. Parlane, A.K. Brown, J.E. Hein, C.P. Berlinguette, Flexible automation accelerates materials discovery, Nature Mater. 21 (7) (2022) 722–726, http://dx.doi.org/10.1038/s41563-021-01156-3.

[9] N. Yachie, R.B. Consortium, T. Natsume, Robotic crowd biology with maholo LabDroids, Nat. Biotechnol. 35 (2017) 310–312, http://dx.doi.org/10.1038/nbt.3758.

[10] G.N. Kanda, T. Tsuzuki, M. Terada, N. Sakai, N. Motozawa, T. Masuda, M. Nishida, C.T. Watanabe, T. Higashi, S.A. Horiguchi, T. Kudo, M. Kamei, G.A. Sunagawa, K. Matsukuma, T. Sakurada, Y. Ozawa, M. Takahashi, K. Takahashi, T. Natsume, Robotic search for optimal cell culture in regenerative medicine, in: S. Méndez-Ferrer, M. Zaidi, S. Méndez-Ferrer, S. Sebastian, C. Monville (Eds.), eLife 11 (2022) e77007, http://dx.doi.org/10.7554/eLife.77007.

[11] M. Terada, Y. Kogawa, Y. Shibata, M. Kitagawa, S. Kato, T. Iida, T. Yorimitsu, A. Kato, K. Matsukuma, T. Maeda, M. Takahashi, G.N. Kanda, Robotic cell processing facility for clinical research of retinal cell therapy, SLAS Technol. 28 (6) (2023) 449–459, http://dx.doi.org/10.1016/j.slast.2023.10.004, URL https://www.sciencedirect.com/science/article/pii/S2472630323000614.

[12] J. Zhang, W. Wan, N. Tanaka, M. Fujita, K. Takahashi, K. Harada, Integrating a pipette into a robot manipulator with uncalibrated vision and TCP for liquid handling, IEEE Trans. Autom. Sci. Eng. (2023) 1–20, http://dx.doi.org/10.1109/TASE.2023.3312657.

[13] B. Burger, P.M. Maffettone, V.V. Gusev, C.M. Aitchison, Y. Bai, X. Wang, X. Li, B.M. Alston, B. Li, R. Clowes, N. Rankin, B. Harris, R.S.S..A.I. Cooper, A mobile robotic chemist, Nature 583 (2020) 237–241, http://dx.doi.org/10.1038/s41586-020-2442-2.

[14] B.J. Choi, W.S. You, S.H. Shin, H. Moon, J.C. Koo, W. Chung, H.R. Choi, Development of robotic laboratory automation platform with intelligent mobile agents for clinical chemistry, in: 2011 IEEE International Conference on Automation Science and Engineering, 2011, pp. 708–713, http://dx.doi.org/10.1109/CASE.2011.6042468.

[15] B.P. MacLeod, F.G.L. Parlane, T.D. Morrissey, F. Hase, L.M. Roch, K.E. Dettelbach, R. Moreira, L.P.E. Yunker, M.B. Rooney, J.R. Deeth, V. Lai, G.J. Ng, H. Situ, R.H. Zhang, M.S. Elliott, T.H. Haley, D.J. Dvorak, A. Aspuru-Guzik, J.E. Hein, C.P. Berlinguette, Self-driving laboratory for accelerated discovery of thin-film materials, Sci. Adv. 6 (20) (2020) eaaz8867, http://dx.doi.org/10.1126/sciadv.aaz8867, arXiv:https://www.science.org/doi/pdf/10.1126/sciadv.aaz8867, URL https://www.science.org/doi/abs/10.1126/sciadv.aaz8867.

[16] Y. Xie, K. Sattari, C. Zhang, J. Lin, Toward autonomous laboratories: Convergence of artificial intelligence and experimental automation, Prog. Mater. Sci. 132 (2023) 101043, http://dx.doi.org/10.1016/j.pmatsci.2022.101043, URL https://www.sciencedirect.com/science/article/pii/S0079642522001244.

[17] Y. Nakajima, M. Hamaya, Y. Suzuki, T. Hawai, F.v. Drigalski, K. Tanaka, Y. Ushiku, K. Ono, Robotic powder grinding with a soft jig for laboratory automation in material science, in: 2022 IEEE/RSJ International Conference on Intelligent Robots and Systems, IROS, 2022, pp. 2320–2326, http://dx.doi.org/10.1109/IROS47612.2022.9981081.

[18] R. Shimizu, S. Kobayashi, Y. Watanabe, Y. Ando, T. Hitosugi, Autonomous materials synthesis by machine learning and robotics, APL Mater. 8 (11) (2020) 111110, http://dx.doi.org/10.1063/5.0020370.

[19] K. Nagai, G.I. Takayuki Osa, T. Tsujiguchi, T. Araki, Y. Kuroda, M. Tomizawa, K. Nagato, Sample-efficient parameter exploration of the powder film drying process using experiment-based Bayesian optimization, Sci. Rep. 12 (2022) 1615, http://dx.doi.org/10.1038/s41598-022-05784-w.

[20] Y. Asano, S. Yoneda, K. Kitai, K. Okada, J. Shiomi, Flexible laboratory automation system based on distributed framework: Implementation for press process in polymer materials development, in: 2023 IEEE 19th International Conference on Automation Science and Engineering, CASE, 2023, pp. 1–6, http://dx.doi.org/10.1109/CASE56687.2023.10260616.

[21] Y. Asano, K. Okada, J. Shiomi, Robotic automation system of polymer press process for materials lab-automation, in: 2024 IEEE/SICE International Symposium on System Integration, SII, 2024, pp. 351–356, http://dx.doi.org/10.1109/SII58957.2024.10417660.
[22] Y. Yataka, A. Suzuki, K. Iijima, M. Hashizume, Enhancement of the mechanical properties of polysaccharide composite films utilizing cellulose nanofibers, Polym. J. 52 (2020) 645–653, http://dx.doi.org/10.1038/s41428-020-0311-3.
[23] Y. Fujikura, M. Kawarai, F. Ozaki, Electrical properties of carbon black-polyethylene composites, Polym. J. 21 (1989) 609–614, http://dx.doi.org/10.1295/polymj.21.609.
[24] M.I. Hussein, S.S. Jehangir, I.J. Rajmohan, Y. Haik, T. Abdulrehman, Q. Clément, N. Vukadinovic, Microwave absorbing properties of metal functionalized-CNT-polymer composite for stealth applications, Sci. Rep. 10 (2020) 16013, http://dx.doi.org/10.1038/s41598-020-72928-1.
[25] Y. Hayashi, J. Shiomi, J. Morikawa, R. Yoshida, Radonpy: automated physical property calculation using all-atom classical molecular dynamics simulations for polymer informatics, npj Comput. Mater. 8 (1) (2022) 222, http://dx.doi.org/10.1038/s41524-022-00906-4.
[26] S. Lo, M. Seifrid, T. Gaudin, A. Aspuru-Guzik, Augmenting polymer datasets by iterative rearrangement, J. Chem. Inf. Mod. 63 (14) (2023) 4266–4276, http://dx.doi.org/10.1021/acs.jcim.3c00144, PMID: 37390494.
[27] S. Takasuka, S. Oikawa, T. Yoshimura, S. Ito, Y. Harashima, T. Takayama, S. Asano, A. Kurosawa, T. Sugawara, M. Hatanaka, T. Miyao, T. Matsubara, Y.-y. Ohnishi, H. Ajiro, M. Fujii, Extrapolation performance improvement by quantum chemical calculations for machine-learning-based predictions of flow-synthesized binary copolymers, Dig. Discover. 2 (2023) 809–818, http://dx.doi.org/10.1039/D2DD00144F, URL http://dx.doi.org/10.1039/D2DD00144F.
[28] A. Wakiuchi, S. Takasuka, S. Asano, R. Hashizume, A. Nag, M. Hatanaka, T. Miyao, Y. Ohnishi, T. Matsubara, T. Ando, T. Sugawara, M. Fujii, H. Ajiro, Composition regulation by flow copolymerization of methyl methacrylate and glycidyl methacrylate with free radical method, Macromol. Mater. Eng. 308 (6) (2023) 2200626, http://dx.doi.org/10.1002/mame.202200626, arXiv: https://onlinelibrary.wiley.com/doi/pdf/10.1002/mame.202200626, URL https://onlinelibrary.wiley.com/doi/abs/10.1002/mame.202200626.
[29] M. Rubens, J. Van Herck, T. Junkers, Automated polymer synthesis platform for integrated conversion targeting based on inline benchtop NMR, ACS Macro Lett. 8 (11) (2019) 1437–1441.
[30] M. Rubens, J.H. Vrijsen, J. Laun, T. Junkers, Precise polymer synthesis by autonomous self-optimizing flow reactors, Ang. Chem. Int. Ed. 58 (10) (2019) 3183–3187, http://dx.doi.org/10.1002/anie.201810384, arXiv: https://onlinelibrary.wiley.com/doi/pdf/10.1002/anie.201810384, URL https://onlinelibrary.wiley.com/doi/abs/10.1002/anie.201810384.
[31] S. Haddadin, S. Parusel, L. Johannsmeier, S. Golz, S. Gabl, F. Walch, M. Sabaghian, C. Jähne, L. Hausperger, S. Haddadin, The franka emika robot: A reference platform for robotics research and education, IEEE Robot. Autom. Mag. 29 (2) (2022) 46–64, http://dx.doi.org/10.1109/MRA.2021.3138382.
[32] http://wiki.ros.org/ROS/Introduction. (Accessed 29 March 2024).
[33] E. Brochu, V.M. Cora, N. de Freitas, A tutorial on Bayesian optimization of expensive cost functions, with application to active user modeling and hierarchical reinforcement learning, 2010, arXiv, abs/1012.2599, URL https://api.semanticscholar.org/CorpusID:1640103.
[34] Y. Motoyama, R. Tamura, K. Yoshimi, K. Terayama, T. Ueno, K. Tsuda, Bayesian optimization package: PHYSBO, Comput. Phys. Comm. 278 (2022) 108405, http://dx.doi.org/10.1016/j.cpc.2022.108405.



Yuki Asano is project lecturer in the Graduate School of Engineering at The University of Tokyo. He received B.E. in Mechanical Engineering from Niigata University in 2007. He received M.A.S. in Interdisciplinary Information Studies from The University of Tokyo in 2013. He received Ph.D. in Mechano-Informatics from The University of Tokyo in 2016. He is a recipient of The Commendation for Science and Technology by the Minister of Education, Culture, Sports, Science and Technology (The Young Scientists' Award). He received Best Paper Award in the 2013 IEEE-RAS International Conference on Humanoid Robots. His research interests include robot system, humanoids and lab-automation.



Kei Okada received BE in Computer Science from Kyoto University in 1997. He received MS and PhD in Information Engineering from The University of Tokyo in 1999 and 2002, respectively. From 2002 to 2006, he joined the Professional Programme for Strategic Software Project in The University of Tokyo. He was appointed as a lecturer in the Creative Informatics at the University of Tokyo in 2006, an associate professor in the Department of Mechano-Informatics in 2009 and a professor in the Department of Mechano-Informatics in 2018. His research interests include humanoids robots, real-time 3D computer vision, and recognition-action integrated system.



Shintaro Nakagawa received his B.S. in 2011, M.S. in 2013, and Ph.D. in 2016 from Tokyo Institute of Technology. From 2016 to 2017, he worked as a JSPS postdoctoral fellow at the Institute for Solid State Physics, the University of Tokyo. In 2017 he was appointed as a research associate at the Institute of Industrial Science, the University of Tokyo, and promoted to a lecturer in 2023. His current research focuses on the synthesis and elucidation of structure–property relationship of homogeneously crosslinked polymers as well as semicrystalline polymers.



Naoko Yoshie received her MS (1990) and PhD (1993) in Polymer Science from Tokyo Institute of Technology. In 1990, she began an academic career as Research Associate at Department of Biomolecular Engineering, Tokyo Institute of Technology, where she actively studied biodegradable and/or biobased polyesters. She joined Institute of Industrial Science, the University of Tokyo as an associate professor in 2001 and promoted to a full professor in 2010, where she expanded her research area to sustainable design of polymers with dynamic bonds, including recycling, self-healing, self-recovering and so on. From 2017 to 2023, she is Chairperson for Polymer Chemistry Subcommittee of Chemistry Committee, Science Council of Japan. She is also engaging in promoting diversity and inclusion of the university as Vice President.



Junichiro Shiomi is Professor in the Institute of Engineering Innovation and Department of Mechanical Engineering, School of Engineering, the University of Tokyo (UTokyo). He received B.E. (1999) from Tohoku University, and Ph. D. (2004) from Royal Institute of Technology (KTH), Sweden. Leading the Thermal Energy Engineering Lab, he has been pursuing research to advance thermal management, waste heat recovery, and energy harvesting technologies based on nano-to-macro innovation in materials, structures, and systems. He is a recipient of the Zeldovich Medal from the Committee on Space Research, the Commendation for Science and Technology by the Minister of Educational, Culture, Sports, Science and Technology (the Young Scientists' Prize and the Science and Technology Prize), the Academic award of Heat Transfer Society of Japan, the Academic Award of Thermoelectric Society of Japan, the JSPS Award, and the Nukiyama Memorial Award.